Anindya Sarkar

Ph.D. Candidate, Computer Science & Engineering
Washington University in St. Louis
Advisor: Prof. Yevgeniy Vorobeychik

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RESEARCH

My goal is to build AI models that can understand, generate, plan, and reason over high-dimensional data across modalities. I am particularly interested in developing generative modeling algorithms that are mathematically grounded and scale to achieve state-of-the-art performance in data-constrained settings, such as scientific discovery and medical diagnosis. My current work aims to improve generative models' search capabilities to accelerate target discovery. I also investigate the capabilities of generative models to tackle planning and reasoning in complex, partially observable environments such as overhead and underwater environments. To tackle this broad range of problems, I draw on tools from Generative Modeling and Sampling, Applied Stochastic Processes and Control, Optimal Transport, and Reinforcement Learning.

Generative Modeling & Sampling — Enhancing inference-time search and fine-grained alignment capabilities of generative models: IMPFM [**Under Review**], BFMT [**Under Review**], BSIT [**Under Review**], PAPA [**ECML 2026**], LATS [**NeurIPS 2026 Workshop on AI for Stochastic Dynamics**]

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Applied Stochastic Process & Control — Active target discovery under partially observable environments via uncertainty-aware generative models: EM-PTDM [**NeurIPS 2025**], DiffATD [**NeurIPS 2025**]

Reinforcement Learning — Visual & Geospatial Active Search for social good: OWL-GPS [**NeurIPS 2026**], VAS [**WACV 2024**], PS-VAS [**NeurIPS 2023**], GOMAA-Geo [**NeurIPS 2024**], DiffVAS [**AAMAS 2026**], AGS [**AAAI 2025, AAMAS 2024**], Mult-Task RL: PACMAN [**ICML 2025**]

Optimal Transport — Out-of-distribution active target discovery via Wasserstein Gradient Flow. [**In Progress**]

Representation Learning — Robustness & Explainability: Ane-hoc Explainability [**CVPR 2022**], Attributional Robustness [**AAAI 2021**], Adversarial Robustness [**NeurIPS 2021**], Robustness against Unseen Noise [**WACV 2022**], Reward-delay attack on RL [**GameSec 2022**]; Graph Neural Network: Representation Capability of Message Passing GNN [**NeurIPS 2022**]

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EDUCATION

Sept 2021 – May 2027 (exp.) — **Ph.D., Computer Science & Engineering**
Washington University in St. Louis, MO — Advisor: Prof. Yevgeniy Vorobeychik

Sept 2021 – June 2023 — **M.S., Computer Science**
Washington University in St. Louis, MO

Sept 2014 – Dec 2016 — **M.Tech., Bio-Medical Engineering**
Indian Institute of Technology Hyderabad, India

RESEARCH EXPERIENCE

Sept 2021 – Present — **Graduate Research Assistant — Washington University in St. Louis**
Enhancing Search Capabilities of Generative Models; Uncertainty-aware generative model for active target discovery in partially observable environments; Generative Models for planning under partially observable environments; Visual & Geo-spatial Active Search. Advisor: Prof. Yevgeniy Vorobeychik.

2020 – 2021 — **Research Assistant — Indian Institute of Technology Hyderabad**
Adversarial machine learning: attributional and adversarial robustness, ante-hoc explainable model architectures, test-time robustness to unseen noise. Advisor: Prof. Vineeth N. Balasubramanian.

July 2016 – Mar 2020 — **R&D Engineer, Deep Learning — Quest Global Pvt. Ltd., India**
Deep learning for computer vision and natural language processing across applied products.

PUBLICATIONS

CONFERENCE PUBLICATIONS

NeurIPS 2026
Adapting Actively on the Fly: Relevance-Guided Online Meta-Learning with Latent Concepts for Geospatial Discovery
Jowaria Khan, **Anindya Sarkar**, Yevgeniy Vorobeychik, Elizabeth Bondi Kelly. arXiv

ECML PKDD 2026
PAPA: Online Personalized Active Preference Alignment
Anindya Sarkar, Nasik Muhammad Nafi, Isaac Lyngaas, Muralikrishnan Gopalakrishnan Meena, Yevgeniy Vorobeychik. arXiv

AAMAS 2026
DiffVAS: Diffusion-Guided Visual Active Search in Partially Observable Environments
Anindya Sarkar, Srikumar Sastry, Aleksis Pirinen, Nathan Jacobs, Yevgeniy Vorobeychik. arXiv

NeurIPS 2025
Active Target Discovery under Uninformative Prior: The Power of Permanent and Transient Memory
Anindya Sarkar*, Binglin Ji*, Yevgeniy Vorobeychik. arXiv

NeurIPS 2025
Online Feedback Efficient Active Target Discovery in Partially Observable Environments
Anindya Sarkar*, Binglin Ji*, Yevgeniy Vorobeychik. arXiv

ICML 2025
Learning Policy Committees for Effective Personalization in MDPs with Diverse Tasks
Luise Ge, Michael Lanier, **Anindya Sarkar**, Bengisu Guresti, Chongjie Zhang, Yevgeniy Vorobeychik. arXiv

AAAI 2025
Active Geospatial Search for Efficient Tenant Eviction Outreach
Anindya Sarkar, Alex DiChristofano, Sanmay Das, Patrick Fowler, Nathan Jacobs, Yevgeniy Vorobeychik. arXiv

NeurIPS 2024
GOMAA-Geo: GOal Modality Agnostic Active Geo-localization
Anindya Sarkar, Srikumar Sastry, Aleksis Pirinen, Chongjie Zhang, Nathan Jacobs, Yevgeniy Vorobeychik. arXiv

WACV 2024
A Visual Active Search Framework for Geospatial Exploration
Anindya Sarkar, Michael Lanier, Scott Alfeld, Jiarui Feng, Roman Garnett, Nathan Jacobs, Yevgeniy Vorobeychik. arXiv

AAMAS 2024
Geospatial Active Search for Preventing Evictions
Anindya Sarkar, Alex DiChristofano, Sanmay Das, Patrick Fowler, Nathan Jacobs, Yevgeniy Vorobeychik. ACM DL

NeurIPS 2023
A Partially-Supervised Reinforcement Learning Framework for Visual Active Search
Anindya Sarkar, Nathan Jacobs, Yevgeniy Vorobeychik. Proceedings

NeurIPS 2022
How Powerful are K-hop Message Passing Graph Neural Networks
Jiarui Feng, Yixin Chen, Fuhai Li, **Anindya Sarkar**, Muhan Zhang. arXiv

CVPR 2022
A Framework for Learning Ante-hoc Explainable Models via Concepts
Anirban Sarkar, Deepak Vijaykeerthy, **Anindya Sarkar**, Vineeth N. Balasubramanian. IEEE

WACV 2022
Leveraging Test-Time Consensus Prediction for Robustness against Unseen Noise
Anindya Sarkar, Anirban Sarkar, Vineeth N. Balasubramanian. CVF

GameSec 2022
Reward Delay Attacks on Deep Reinforcement Learning
Anindya Sarkar, Jiarui Feng, Yevgeniy Vorobeychik, Christopher Gill, Ning Zhang. arXiv

NeurIPS 2021
Get Fooled for the Right Reason: Improving Adversarial Robustness through a Teacher-Guided Curriculum Learning Approach
Anindya Sarkar, Anirban Sarkar, Sowrya Gali, Vineeth N. Balasubramanian. arXiv

AAAI 2021
Enhanced Regularizers for Attributional Robustness
Anindya Sarkar, Anirban Sarkar, Vineeth N. Balasubramanian. AAAI

ICANN 2020
Enforcing Linearity in DNN Succours Robustness and Adversarial Image Generation
Anindya Sarkar, Raghu Iyengar. Springer

ICMLA 2020
ODE-Guided Neural Data Augmentation Techniques for Time Series Data and its Benefits on Robustness
Anindya Sarkar, Anirudh Raj, Raghu Iyengar. arXiv

NLPIR 2019
Zero-Shot Multilingual Sentiment Analysis using Hierarchical Attentive Network and BERT
Anindya Sarkar, Sujeeth Reddy, Raghu Iyengar. ACM DL

MANUSCRIPTS UNDER REVIEW

arXiv 2026
Sampling Meets Interaction: Sequentially-Controlled Multi-Particle Flow-Maps for Efficient Inference-time Search
Binglin Ji*, **Anindya Sarkar***, Hengchang Lu, Jens Sjölund, Yevgeniy Vorobeychik. *arXiv:2607.01144*. arXiv

arXiv 2026
BFMT: Enhancing Search Capabilities of Tree Sampler via Bootstrap Flow-Map Tree
Binglin Ji*, **Anindya Sarkar***, Hengchang Lu, Kunyu Wang, Alessandro Tibo, Jens Sjölund, Yevgeniy Vorobeychik. *arXiv:2607.02915*. arXiv

Under review
BSIT: Bootstrap Stochastic Interpolant Tree for Efficient Search under Sparse Reward
Binglin Ji*, **Anindya Sarkar***, Yevgeniy Vorobeychik.

WORKSHOP PUBLICATIONS

NeurIPS Workshop 2026
LATS: Levy Adaptive Tree Sampling for Feedback-Driven Diverse Target Discovery
Binglin Ji*, **Anindya Sarkar***, Hengchang Lu, Lecheng Kong, Yixin Chen, Yevgeniy Vorobeychik. *AI for Stochastic Dynamics*. arXiv

AAAI Workshop 2024
Attacks on Node Attributes in Graph Neural Networks
Xu Ying, Michael Lanier, **Anindya Sarkar**, Yevgeniy Vorobeychik. *AI for Cyber Security*. arXiv

ECCV Workshop 2022
Empowering a Robust Model with Stable and Object-Aligned Explanations
Sowrya Gali, **Anindya Sarkar**, Vineeth N. Balasubramanian. *Adversarial Robustness in the Real World*. PDF

ECCV Workshop 2020
Inducing Semantic Grouping of Latent Concepts for Explanations: An Ante-Hoc Approach
Anirban Sarkar, Deepak Vijaykeerthy, **Anindya Sarkar**, Vineeth N. Balasubramanian. arXiv

JOURNAL PUBLICATIONS

Front. Neuroinf. 2019
Curated Model Development Using NEUROiD: A Web-Based NEUROmotor Integration and Design Platform
Anindya Sarkar. *Frontiers in Neuroinformatics*. Link

TALKS AND PRESENTATIONS

Oct 2025
Enhancing Exploration Capabilities of Generative Models via Interactive Particle Dynamics
Invited talk, AstraZeneca, Gothenburg, Sweden (Molecular AI, Discovery Sciences) — hosted by Dr. Alessandro Tibo

Oct 2025
Visual Active Search for Social Good
Invited talk, University of Michigan — hosted by Prof. Elizabeth Bondi-Kelly

May 2024 — **Doctoral Consortium — AAMAS 2024**
23rd International Conference on Autonomous Agents and Multiagent Systems

Conference presentations — Presented work at **NeurIPS** (2021, 2023, 2024, 2025 ×2), **AAAI** (2021, 2025), **AAMAS** (2024, 2026), **WACV** (2022, 2024), **ECML PKDD** (2026), **GameSec** (2022), **ICANN** (2020), **ICMLA** (2020), and **NLPIR** (2019).

TEACHING

Fall 2024 — **Teaching Assistant — CS755, Adversarial Machine Learning**
Computer Science & Engineering, Washington University in St. Louis

MENTORING

2025 – 2026 — **Binglin Ji — M.S., Washington University in St. Louis**
Two NeurIPS 2025 papers (DiffATD; EM-PTDM), and one NeurIPS 2026 workshop (LATS) and three manuscripts under review on the search capabilities of generative models.

2026 — **Xinyu Wang — M.S., Washington University in St. Louis**
Safe active search in underwater environments (in progress).

2025 – 2026 — **Hengchang Lu — M.S., Northwestern University**
One NeurIPS 2026 workshop (LATS) and three manuscripts under review on the search capabilities of generative models.

2021 — **Sowrya Gali — Undergraduate, Indian Institute of Technology Hyderabad**
[NeurIPS 2021] paper on teacher-guided curriculum learning for adversarial robustness; [ECCV Workshop 2022] paper on object-aligned robust explanation of neural network.

PROFESSIONAL SERVICE

Program committee — **AAAI** — Program Committee member, 2024–2026.

Conference reviewer — **NeurIPS** (2021–2026), **ICML** (2021–2026), **CVPR** (2022–2026), **AAMAS** (2022–2026), **ICLR** (2023–2026), **ECCV** (2024–2026).

HONORS AND AWARDS

2024-2026 — **Departmental Honors — top 15% of Ph.D. students**
Department of Computer Science & Engineering, Washington University in St. Louis

2024 — **AAMAS Student Scholarship**
AAMAS Conference Committee — conference attendance and Doctoral Consortium

2014-2016 — **MHRD Scholarship**
Ministry of Human Resource Development, Government of India — for M.Tech. study

TECHNICAL SKILLS

Software — Python, C/C++, MATLAB; PyTorch, JAX, NumPy/SciPy, Hugging Face.

Infrastructure — Distributed multi-GPU training, SLURM HPC clusters.

REFERENCES

Ph.D. advisor — **Prof. Yevgeniy Vorobeychik**
Computer Science & Engineering, Washington University in St. Louis

Former advisor — **Prof. Vineeth N. Balasubramanian**
Computer Science & Engineering, Indian Institute of Technology Hyderabad

Collaborator — **Prof. Jens Sjölund**
Computer Science & Engineering, Uppsala University, Sweden